정답 (Answers)

7.9.1 단답형

1. a. 6Ah b. EAh c. FDh d. A9h

2. a. 9Ah b. 6Ah c. A9h (CF=1) d. 3Ah (CF=1)

3. DX = 0002h, AX = 2200h

4. AX = 0306h (AL=06h 몫, AH=03h 나머지)

5. EAX = 00012340h, EDX = 00000000h

6. 나눗셈 오버플로 예외 발생 (몫 500400h가 16비트를 초과)

7. BX = 0066h

8. 나눗셈 오버플로 예외 발생 (RDX 108h ≥ 제수 100h), 출력 없음

9.

```
오류 4가지: ① OFFSET 누락 ② cx → ecx ③ dec → inc ④ 결과를 result에 저장

mov  ecx,8
mov  esi,OFFSET val1
mov  edi,OFFSET val2
mov  ebx,OFFSET result
clc
top:
mov  al,[esi]
sbb  al,[edi]
mov  [ebx],al
inc  esi
inc  edi
inc  ebx
loop top
; result = 1AEE1FD3EBFA16EFh
```

10. RAX = 0004080C10140000h

7.9.2 알고리즘 워크벤치

1.

```
shl eax,16
sar eax,16
```

2.

```
shr al,1
jnc done
or  al,80h
done:
```

3.

```
shl eax,4
```

4.

```
shr ebx,2
```

5.

```
rol dl,4
```

6.

```
shld dx,ax,1
```

7.

```
shr byteArray+2,1
rcr byteArray+1,1
rcr byteArray,1
; 결과: 40h, 90h, 19h
```

8.

```
shl wordArray,1
rcl wordArray+2,1
rcl wordArray+4,1
; 결과: 021Ah, 80C9h, 2757h
```

9.

```
mov  ax,-5
mov  bx,3
imul bx
mov  val1,ax        ; -15 (FFF1h)
```

10.

```
mov  ax,-276
cwd
mov  bx,10
idiv bx
mov  val1,ax        ; 몫 -27 (FFE5h), 나머지 DX = -6
```

11.

```
mov eax,val2
mul val3
mov ebx,val4
sub ebx,3
div ebx
mov val1,eax
```

12.

```
mov  eax,val2
cdq
idiv val3
mov  ebx,val1
add  ebx,val2
imul ebx
mov  val1,eax
```

13.

```
showDecimal8 PROC
    mov  ah,0
    mov  bl,10
    div  bl
    or   ax,3030h
    push eax
    call WriteChar
    pop  eax
    mov  al,ah
    call WriteChar
    ret
showDecimal8 ENDP
```

14. AX = 0108h (AF = CF = 1)

15.

```
mov eax,y
sub eax,1
and eax,n
mov x,eax
```

16.

```
mov edx,eax
sar edx,31
add eax,edx
xor eax,edx
```